

momentum - practice problems

name _____

1. A 160 g hockey puck slides across the ice at 30 m/s after a slap shot. What is its momentum?
2. A 0.060 kg tennis ball is moving left at 25 m/s just after leaving a racket. What is the ball's momentum?
3. Which has more momentum: a 2000 kg car rolling through a parking lot at 1.0 m/s, or a 70 kg runner sprinting at 9.0 m/s?

4. A 70 kg runner has 420 kg m/s of momentum. How fast is the runner moving?

5. A fan pushes a 2.0 kg dynamics cart, and the cart speeds up from 1.0 m/s to 5.0 m/s. What impulse did the fan give the cart?

6. A student pushes a box with an average force of 75 N for 0.20 s. What impulse does the student give the box?

7. A soccer player kicks a 0.45 kg ball from rest, and it leaves the player's foot at 25 m/s. What average force does the foot apply to the ball?

8. A 0.60 kg basketball hits the floor moving down at 6.0 m/s and bounces back up at 4.5 m/s. The ball touches the floor for 0.020 s. What is the average net force on the ball during the bounce?

9. A 1200 kg car (including its passengers) slows from 72 km/h to rest. The average braking force is 6000 N opposite the motion. How long does it take to stop?

10. A pitcher throws a 0.145 kg baseball at 90 miles per hour. What is the ball's momentum? (1 mile = 1609 m)

11. A 5.0 kg cart starts from rest. A steady 10 N net force pushes it for 3.0 s. What is the cart's final velocity?

12. A 3.0 kg remote control boat starts from rest and accelerates at 2.0 m/s^2 for 4.0 s. What is its momentum at the end?

13. An egg dropped onto a pillow survives, but the same egg dropped from the same height onto a hard floor breaks. Both eggs stop, so they have the same change in momentum. Why does only one break?

14. In a crash test, a 75 kg driver slows from 54 km/h to rest. Compare the average force on the driver if the stopping time is 0.10 s with the average force if an airbag and crumple zone stretch the stopping time to 0.60 s.